

PRAMOD KUMAR MAHATO

AI / Machine Learning Engineer | Full Stack Developer (Python, FastAPI, LangChain, RAG & Multi-Agent Systems)

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PROFESSIONAL SUMMARY

AI/ML and Full Stack Engineer with hands-on experience building production-oriented backend and generative-AI systems using Python, FastAPI, LangChain, and LangGraph. Skilled in designing multi-agent architectures, RAG pipelines, and vector-search applications with FAISS, Groq, and PostgreSQL, and in deploying containerized services with Docker. Proven ability to cut manual effort by 70%+ and deliver context-aware, tool-integrated AI solutions with measurable business impact.

TECHNICAL SKILLS

AI / LLM Engineering: LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Supervisor Agents, AI Guardrails, Human-in-the-Loop (HITL), Model Context Protocol (MCP), LangSmith, Prompt Engineering, Groq, FAISS, Pinecone, Sentence Transformers

Backend & APIs: Python, FastAPI, REST API Design, Microservices, Async Programming, Streamlit

Data & Storage: PostgreSQL, MySQL, MongoDB, Redis, FAISS (Vector DB), Pinecone (Vector DB), Data Ingestion Pipelines

Cloud & DevOps: Docker, CI/CD, GitHub Actions, Git, Render, Cloud Deployment

Frontend: JavaScript, React.js, Streamlit (UI dashboards for AI applications)

Integrations & Tools: Tavily, AviationStack, MVC Architecture, RBAC

PROFESSIONAL EXPERIENCE

Full Stack Developer Intern — Excellence Technologies Pvt. Ltd.

Apr 2026 – Jul 2026

Noida, Sector 8, India | 4 Months

- Built and deployed full-stack applications using Python, FastAPI, JavaScript, and Streamlit, implementing REST APIs, frontend integration, and debugging workflows as part of real-world software engineering practices.
- Engineered backend and AI-driven applications using LangChain, LangGraph, Groq, FAISS, PostgreSQL, MCP, and Docker, strengthening a production-oriented foundation for AI/ML system design and cloud deployment.
- Delivered an HR Policy RAG Assistant by implementing document processing, embeddings, vector search, and RAG pipelines with LangChain, Groq, FAISS, FastAPI, and Streamlit, enabling context-aware document Q&A.
- Designed TripMate AI, a multi-agent travel-planning system with LangGraph, MCP, Supervisor Agents, AI Guardrails, and HITL workflows, gaining practical exposure to agent orchestration, tool integration, observability, and deployment.

PROJECTS

TripMate AI — Multi-Agent Travel Planner | GitHub

Python · FastAPI · LangGraph · LangChain · Groq · MCP · PostgreSQL · Tavily · AviationStack · HITL · AI Guardrails · Docker

- Architected a multi-agent travel planning system with 4+ specialized agents for flight research, hotel discovery, itinerary generation, and final response synthesis, orchestrated through LangGraph and exposed through a FastAPI backend.
- Implemented Model Context Protocol (MCP) with custom server/client components and a Supervisor Agent with AI Guardrails and HITL approval workflow, connecting agents to Tavily and AviationStack for real-time travel research and flight data.
- Dockerized the application and implemented persistent conversation state using PostgreSQL, creating a production-oriented architecture with 5+ integrated AI/tooling components and end-to-end API-driven travel planning.

HR Policy RAG Assistant | GitHub

Python · LangChain · Groq · FAISS · Sentence Transformers · Streamlit · RAG

- Engineered an end-to-end RAG pipeline with 6+ processing stages, including document ingestion, text chunking, embeddings, FAISS vector indexing, semantic retrieval, and context-aware LLM generation for HR policy Q&A.
- Implemented FAISS-based semantic search and Sentence Transformer embeddings, enabling retrieval from 100% of indexed policy content and improving contextual response relevance through optimized chunking and retrieval strategies.
- Integrated LangChain with Groq LLMs and developed a Streamlit interface, reducing manual HR policy lookup effort by an estimated 70%+ and achieving ~90%+ response relevance across manually evaluated queries.

EDUCATION

B-Tech, Computer Science and Engineering (Specialization in AI & ML)

Aug 2022 – Jul 2026

Uttarakhand Technical University, Dehradun | CGPA: 7.0 / 10.0

12th (CBSE) — DAV Public School, Kusunda, Jharkhand | 80.8%

June 2021

10th (ICSE) — De Nobili School, Koradih, Jharkhand | 83%

Mar 2019

ACHIEVEMENTS

- Won 1st Runner-Up among 50+ teams at a Roorkee Institute of Technology hackathon (Jan 2025); also competed at HackIndia.